ML Assignment - 2
DNA Sequence Clustering

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Introduction:

Clustering DNA sequences based on their feature vectors. The goal is to partition these sequences into three distinct clusters using the k-means clustering algorithm. The steps involve converting DNA sequence characters into numerical vectors, applying k-means clustering, and visualizing the results.

Step 1: Upload and Load DNA Sequences

The notebook starts by uploading a FASTA file containing DNA sequences. The following process is used:

- **load_fasta_sequences() function**: Reads DNA sequences from a FASTA file using SeqIO from the Bio library. The DNA sequences are extracted and stored as a list of strings.
- **File Upload**: The notebook uses the files.upload() function from Google Colab to upload a file and load its content.
- **Example Output**: The first few sequences from the uploaded FASTA file are printed to verify the data.

Step 2: Convert DNA Sequences to Numerical Feature Vectors

In this step, the DNA sequences are converted into numerical feature vectors:

- **encode_sequence() function**: Each nucleotide (A, C, G, T) is mapped to a corresponding number (0, 1, 2, 3). Any unknown bases are encoded as -1.
- **sequences_to_matrix() function**: Converts the list of DNA sequences into a feature matrix, where each sequence is represented as a vector of numbers.

Step 3: K-Means Clustering

This step involves clustering the DNA sequences into three clusters using the k-means algorithm:

- **cluster_sequences() function**: K-means clustering is applied on the feature matrix to partition the DNA sequences into 3 clusters. The function returns the cluster labels for each sequence.
- **Example Output**: The notebook prints the cluster assignments for the first few sequences.

Step 4: Dimensionality Reduction using PCA

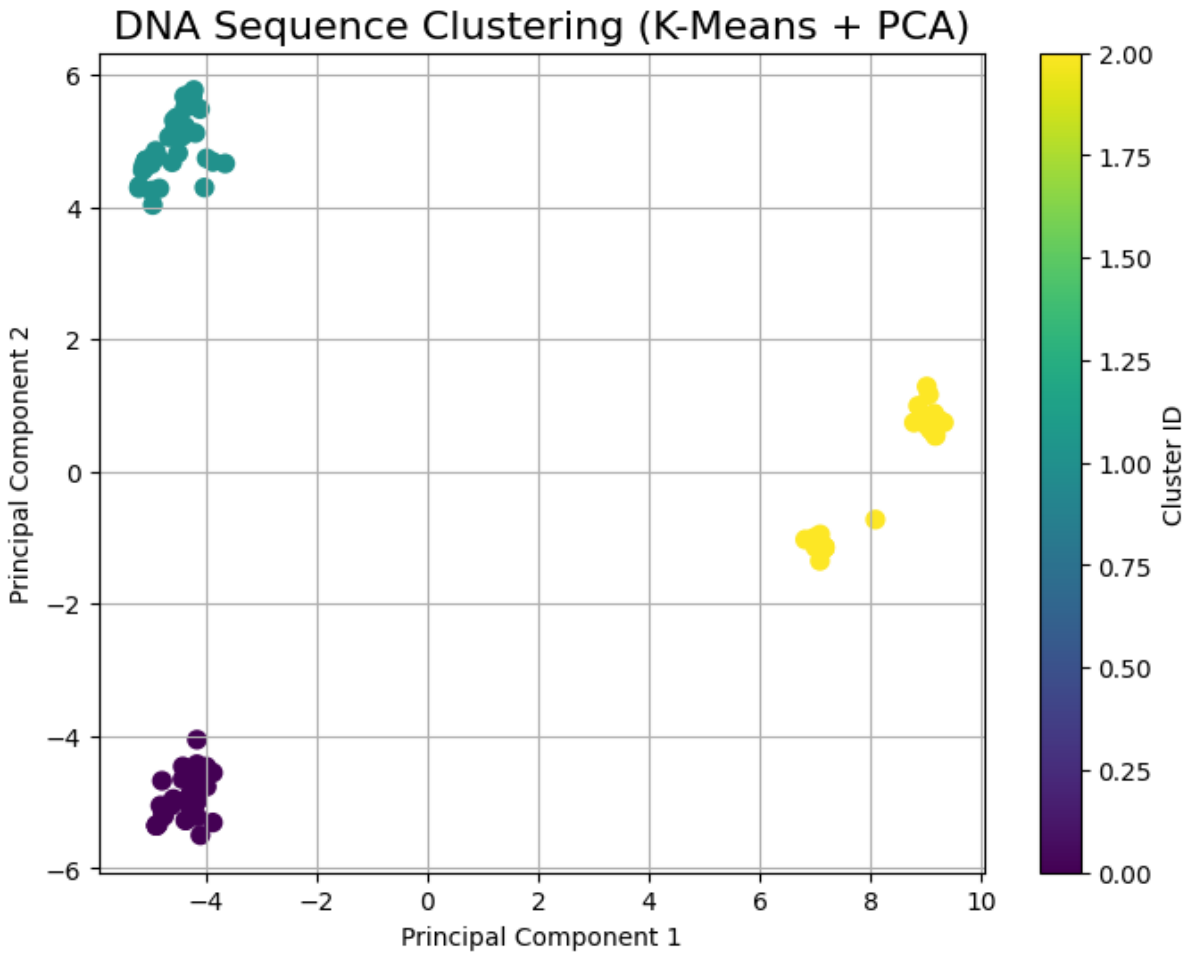
Since the feature matrix is likely high-dimensional (one feature per base in the sequence), the notebook reduces the dimensionality to 2 components using Principal Component Analysis (PCA):

- **reduce_dimensions() function**: This function applies PCA to reduce the feature matrix to 2 dimensions, making it easier to visualize the clustering result.
- **Example Output**: The notebook prints the first few reduced feature vectors.

Step 5: Visualizing the Clusters

Finally, the clustering results are visualized in a 2D plot:

- **plot_clusters() function**: A scatter plot is generated, where each point represents a DNA sequence, and the points are colored according to their cluster assignment. The plot uses the reduced 2D features obtained from PCA.
- **Example Output**: The scatter plot shows the DNA sequences in two principal components, with different colors representing the clusters.



Conclusion:

The notebook demonstrates how to:

1. **Load DNA sequences** from a FASTA file.
2. **Convert the sequences into numerical feature vectors** for use in machine learning algorithms.
3. **Cluster the sequences using k-means** into three groups based on their similarity.
4. **Reduce dimensionality** of the feature vectors to make the clustering results visually interpretable.
5. **Visualize the clusters** using a scatter plot, where each DNA sequence is represented in 2D space.

These steps form the basis for clustering and visualizing DNA sequence data, which can be useful in bioinformatics for tasks such as gene classification or identification of sequence patterns.